Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
#include<stdio.h>
void func1(int *arr){
  for(int i=1; i<=3;i++)
    (i-1)[arr] = (i%3);
}
void func2 (int a[1] , int arr){
  a[0] = arr + 1;
}
int main(){
  int arr[4] = {0,0,0,0};
  func1(arr+1);
  for(int i=0; i<4; i++)
        printf("%d ", i[arr] );
  printf("\n");
  func2(arr, *arr);
  for(int i=0; i<4; i++)
        printf("%d ", i[arr] );
  return 0;
}
```

Output
1200
2200

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
 int temp = arr + 1;
 a = &temp;
}
```

Explain your answer

Yes, it will change as a and a[0] = arr+1 will point at the same address only for the 1st element and will be different for rest terms.

Q. 2: Write the output of the following sequence of instructions **long arr2[100]; int arr1[100], a1=(&arr2[50]), a2=(&arr2[10]), b1=(&arr1[50]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

320, 160

Here, a1, a2, b1, b2, represent a specific address of arr1 and arr2

Each arr1[i] (int) take 4 bits to store data while arr2[i] (long) each takes 8 bits to store data. ∴ $a_1 - a_2 = 8(50-10) = 320$

$b_1 - b_2 = 4(50-10) = 160$